Topic 4: Analysis Under the Randomized Block Design Chapters 11.5-11.6 and 15.7 of our Textbook

STAT 2509: Introduction to Statistical Modeling II

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Motivation for Blocking

- In completely randomized designs (CRD), all variation besides treatment effects goes into error.
- **Blocking**: Group experimental units into homogeneous blocks to control for known nuisance variables (e.g., soil fertility, age, machine, operator).
- Each **treatment** appears once in each block $\rightarrow$ reduces unexplained variability.
- **Example**: Testing 4 fertilizer types (treatments) on plots; each farm is a block with varying soil quality.

RBD Notation and Layout

Structure

- k treatments: $T_1, T_2, \ldots, T_k$
- b blocks: $B_1, B_2, \ldots, B_b$
- One observation per treatment-block combination: y_{ij} = observation for treatment i in block j
- Total observations: $n = k \times b$

Table: RBD Data Layout

	B_1	B_2	$\cdots$	B_b	**Treatment Totals**
T_1	y_{11}	y_{12}	$\cdots$	y_{1b}	$T_1 = \sum_{j=1}^{b} y_{1j}$
T_2	y_{21}	y_{22}	$\cdots$	y_{2b}	$T_2 = \sum_{j=1}^{b} y_{2j}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
T_k	y_{k1}	y_{k2}	$\cdots$	y_{kb}	$T_k = \sum_{j=1}^{b} y_{kj}$
Block Totals	$B_1 = \sum_{i=1}^{k} y_{i1}$	$B_2 = \sum_{i=1}^{k} y_{i2}$	$\cdots$	$B_b = \sum_{i=1}^{k} y_{ib}$	$G.T. = \sum_{i=1}^{k} \sum_{j=1}^{b} y_{ij}$

Model and Assumptions

Model:

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$$

where:

- μ = overall mean
- τ_i = effect of treatment i ($\sum_{i=1}^{k} \tau_i = 0$)
- β_j = effect of block j ($\sum_{j=1}^{b} \beta_j = 0$)
- $\varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$

Assumptions:

1. k randomly selected units from each block are assigned randomly to the treatments, with one unit in each treatment. (In other words, the observations are independent across blocks and collected using a randomized block design, RBD.)
2. Within each block, errors are normally distributed. (In other words, all treatment-block populations are normally distributed.)
3. Constant variance across all treatment-block combinations.
4. Additivity: There is no interaction between treatments and blocks.

Check Constant Variance In Assumption 3

To check equal variance in assumption 3), use **Hartley's test** for checking constant variance across treatment populations as done for CRD.

$H_0 : \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2,$ where σ_i^2's are the variances of the treatment populations

$H_a :$ Not all σ_i^2's are the same

Test statistic:

$$F_{max} = \frac{S_{max}^2}{S_{min}^2}$$

where S_i^2's are the sample variances of the treatment samples.

Reject H_0 if $F_{max} > F_{max}(k, [n] - 1); \alpha$

If H_0 is not rejected (then assumption (3) is ok), proceed with the main hypothesis of interest.

Total variation partitioned into:

$$\text{Total sum of squares: } TSS = \sum_{i=1}^{k}\sum_{j=1}^{b} y_{ij}^2 - \frac{1}{bk}\left(\sum_{i=1}^{k}\sum_{j=1}^{b} y_{ij}\right)^2$$

$$= \sum_{i=1}^{k}\sum_{j=1}^{b} y_{ij}^2 - \frac{(G.T.)^2}{n}$$

$$\text{Sum of squares for treatments: } SSTr = \sum_{i=1}^{k}\frac{T_i^2}{b} - \frac{(G.T.)^2}{n}, \text{ where } T_i = \sum_{j=1}^{b} y_{ij}, i = 1, \cdots, k$$

$$\text{Sum of squares for blocks: } SSB = \sum_{j=1}^{b}\frac{B_j^2}{k} - \frac{(G.T.)^2}{n}, \text{ where } B_j = \sum_{i=1}^{k} y_{ij}, j = 1, \cdots, b$$

$$\text{Sum of squares for errors: } SSE = TSS - SSTr - SSB$$

$$\overline{T_i} = \frac{T_i}{b}, \quad \overline{B_j} = \frac{B_j}{k}, \quad n = bk \text{ (total sample size)}; \quad d.f._{TSS} = n - 1 = bk - 1$$

$$d.f._{SSTr} = k - 1 \quad \Rightarrow \quad MSTr = \frac{SSTr}{k-1}$$

$$d.f._{SSB} = b - 1 \quad \Rightarrow \quad MSB = \frac{SSB}{b-1}$$

$$d.f._{SSE} = bk - k - b + 1 = (b-1)(k-1) \quad \Rightarrow \quad MSE = SSE/(bk - b - k + 1)$$

ANOVA Table for RBD

Table: RBD ANOVA Table

Source	df	SS	MS	F
Treatments	$k-1$	SSTr	MSTr = $\frac{\text{SSTr}}{k-1}$	$F_{\text{tr}} = \frac{\text{MSTr}}{\text{MSE}}$
Blocks	$b-1$	SSB	MSB = $\frac{\text{SSB}}{b-1}$	$F_{\text{block}} = \frac{\text{MSB}}{\text{MSE}}$
Error	$(k-1)(b-1)$	SSE	MSE = $\frac{\text{SSE}}{(k-1)(b-1)}$	
Total	$n-1$	TSS		

Hypothesis Testing

For treatments

$$H_0 : \tau_1 = \tau_2 = \cdots = \tau_k = 0 \text{ (No treatment effects)}$$
$$H_a : \text{ At least one } \tau_i \neq 0$$

equivalently,

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \text{ (No treatment effects)}$$
$$H_a : \text{ At least one } \mu_i \neq \mu_j \text{ for some } i \neq j.$$

Test statistic:

$$F_{\text{tr}} = \frac{\text{MSTr}}{\text{MSE}} \sim F_{k-1,(k-1)(b-1)} \text{ under } H_0$$

Reject H_0 if $F_{\text{tr}} > F_{\alpha,k-1,(k-1)(b-1)}$.

For blocks (usually not of primary interest)

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_b = 0$$

Test statistic:

$$F_{\text{block}} = \frac{\text{MSB}}{\text{MSE}}$$

Significant block effect $\Rightarrow$ blocking was effective.

Example: Crop Yield with 3 Fertilizers, 4 Farms

A agricultural research team wants to compare the effectiveness of three different fertilizer formulations (Fertilizer A, B, and C) on crop yield. They suspect that soil quality varies significantly across different farms, which could obscure the true fertilizer effects if not controlled for. To address this, they implement a Randomized Block Design (RBD).

Table: Yield (kg/plot) for $k = 3$ fertilizers in $b = 4$ farms

	Farm 1	Farm 2	Farm 3	Farm 4	T_i
Fertilizer A	12	14	11	13	50
Fertilizer B	15	16	14	17	62
Fertilizer C	10	12	9	11	42
B_j	37	42	34	41	$G.T. = 154$

Step 1: Calculate intermediate quantities

- $n = bk = 3 \times 4 = 12$
- $\sum\sum y_{ij}^2 = 12^2 + 14^2 + 11^2 + 13^2 + 15^2 + 16^2 + 14^2 + 17^2 + 10^2 + 12^2 + 9^2 + 11^2 = 2046$
- $\sum T_i^2/b = (50^2 + 62^2 + 42^2)/4 = (2500 + 3844 + 1764)/4 = 8108/4 = 2027$
- $\sum B_j^2/k = (37^2 + 42^2 + 34^2 + 41^2)/3 = (1369 + 1764 + 1156 + 1681)/3 = 5970/3 = 1990$
- $G.T.^2/n = 154^2/12 = 23716/12 = 1976.333$

Example: Sum of Squares Calculations

Using the formulas:

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{b} y_{ij}^2 - \frac{(G.T.)^2}{n} = 2046 - 1976.333 = 69.667$$

$$SSTr = \sum_{i=1}^{k} \frac{T_i^2}{b} - \frac{(G.T.)^2}{n} = 2027 - 1976.333 = 50.667$$

$$SSB = \sum_{j=1}^{b} \frac{B_j^2}{k} - \frac{(G.T.)^2}{n} = 1990 - 1976.333 = 13.667$$

$$SSE = TSS - SSTr - SSB = 69.667 - 50.667 - 13.667 = 5.333$$

Proportion of variation explained:

- Treatments explain: $50.667/69.667 = 72.7\%$ of total variation
- Blocks explain: $13.667/69.667 = 19.6\%$ of total variation
- Unexplained error: $5.333/69.667 = 7.7\%$

Example: Mean Squares and F-tests

Degrees of freedom:

$$d.f.TSS = n - 1 = 12 - 1 = 11$$
$$d.f.SSTr = k - 1 = 3 - 1 = 2$$
$$d.f.SSB = b - 1 = 4 - 1 = 3$$
$$d.f.SSE = (k-1)(b-1) = 2 \times 3 = 6$$

Calculate Mean Squares:

$$MSTr = \frac{SSTr}{k-1} = \frac{50.667}{2} = 25.333$$
$$MSB = \frac{SSB}{b-1} = \frac{13.667}{3} = 4.556$$
$$MSE = \frac{SSE}{(k-1)(b-1)} = \frac{5.333}{6} = 0.889$$

Calculate F-statistics:

$$F_{\text{tr}} = \frac{MSTr}{MSE} = \frac{25.333}{0.889} = 28.5$$
$$F_{\text{block}} = \frac{MSB}{MSE} = \frac{4.556}{0.889} = 5.125$$

Example: ANOVA Table and Conclusion

Table: ANOVA for Fertilizer Example

Source	df	SS	MS	F
Fertilizer (Treatment)	2	50.667	25.333	28.5
Farm (Block)	3	13.667	4.556	5.125
Error	6	5.333	0.889	
Total	11	69.667		

Critical values and conclusions:

- For treatments: $F_{0.05,2,6} = 5.14$
- Since $28.5 > 5.14$, reject H_0: **Significant fertilizer effects exist**
- For blocks: $F_{0.05,3,6} = 4.76$
- Since $5.125 > 4.76$, reject H_0: **Blocking was effective**

Practical conclusions:

- Fertilizer B gives highest average yield: $62/4 = 15.5$ kg/plot
- Fertilizer A gives intermediate yield: $50/4 = 12.5$ kg/plot
- Fertilizer C gives lowest yield: $42/4 = 10.5$ kg/plot
- Farm 2 is most productive: $42/3 = 14.0$ kg/plot average
- Farm 3 is least productive: $34/3 = 11.333$ kg/plot average

Further Comparison of RBD and CRD: Relative Efficiency

Primary Goal of Both Designs

- Compare treatment means
- Both use sample means as estimators:
  - RBD: $\overline{T_i} = \dfrac{T_{i.}}{b}$
  - CRD: $\overline{y_i} = \dfrac{\sum \text{observations for treatment } i}{r}$
- Both estimators are **unbiased**: $E(\overline{T_i}) = \mu_i$ and $E(\overline{y_i}) = \mu_i$

Key Question

"Which design gives more precise estimates of treatment means?"

Precision measured by variance: $\text{Var}(\overline{T_i})$ vs. $\text{Var}(\overline{y_i})$

Practical Implication

To achieve the **same estimation precision**, which design requires **fewer observations**?

Smaller sample size = Lower cost of data collection

Estimating Variances from RBD Data

The true variances $\text{Var}(\overline{T_i})$ (RBD) and $\text{Var}(\overline{y_i})$ (CRD) are unknown parameters.

Unbiased Estimators from RBD ANOVA Results

1. **For RBD variance**:
$$\widehat{Var}(\overline{T}_i) = \frac{MSE_{RB}}{b}$$
where MSE_{RB} is the Mean Square Error from RBD ANOVA. text
2. **For CRD variance** (assuming equal sample sizes $n_1 = n_2 = \cdots = n_k = r$):
$$\widehat{\text{Var}}(\overline{y_i}) = \frac{(b-1)\text{MSB} + b(k-1)\text{MSE}_{\text{RB}}}{(bk-1)r}$$
where MSB is the Mean Square for Blocks from RBD ANOVA.

When Are RBD and CRD Equally Precise?

If $Var(\overline{T}_i) = Var(\overline{y}_i)$ (same estimation precision), then their estimators should be approximately equal:

$$\widehat{Var}(\overline{T}_i) \approx \widehat{Var}(\overline{y}_i)$$

- **Substitute the estimators:**

$$\frac{MSE_{RB}}{b} \approx \frac{(b-1)MSB + b(k-1)MSE_{RB}}{(bk-1)r}$$

- **Solve for r/b ratio:**

$$\frac{r}{b} \approx \frac{(b-1)MSB + b(k-1)MSE_{RB}}{(bk-1)MSE_{RB}}$$

where:

- MSB = Mean Square for Blocks from RBD ANOVA
- MSE_{RB} = Mean Square Error from RBD ANOVA
- b = number of blocks
- k = number of treatments

Interpreting the Ratio r/b

Recall:

- r = treatment sample size for CRD
- b = treatment sample size for RBD
- Ratio r/b = how many times more observations per treatment needed in CRD

The Ratio Formula:

$$R.E. = \frac{r}{b} \approx \frac{(b-1)MSB + b(k-1)MSE_{RB}}{(bk-1)MSE_{RB}}$$

Interpretation:

To achieve the **same estimation precision** for CRD and RBD, approximately

$$\frac{(b-1)MSB + b(k-1)MSE_{RB}}{(bk-1)MSE_{RB}}$$

times as many observations in each treatment group would be needed if CRD is used as those needed for RBD.

Interpreting R.E. Values

Case 1: R.E. > 1

- Larger treatment sample size needed for CRD
- **RBD is more efficient than CRD**
- Example: R.E. = 1.5
  - CRD needs 1.5× more observations
  - If RBD needs 10 obs/treatment
  - CRD needs 15 obs/treatment

Case 2: R.E. < 1

- Smaller treatment sample size needed for CRD
- **RBD is less efficient than CRD**
- Example: R.E. = 0.8
  - CRD needs only 80% of observations
  - If RBD needs 10 obs/treatment
  - CRD needs 8 obs/treatment

Case 3: R.E. = 1

- Both designs require same sample size
- RBD and CRD are equally efficient

Numerical Example

Given Parameters:

- $b = 4$ (4 blocks), and $k = 3$ (3 treatments)
- From RBD ANOVA table: $\text{MSB} = 7.316$ and $\text{MSE}_{\text{RB}} = 2.885$

Calculate R.E.:

$$
\begin{aligned}
R.E. &= \frac{(b-1)\text{MSB} + b(k-1)\text{MSE}_{\text{RB}}}{(bk-1)\text{MSE}_{\text{RB}}} \\
&= \frac{(4-1)(7.316) + 4(3-1)(2.885)}{(4 \times 3 - 1)(2.885)} = \frac{45.028}{31.735} \\
&\approx 1.419
\end{aligned}
$$

Interpretation:

To achieve the **same precision** for comparing treatment means:

- CRD would need approximately **1.419 times** as many observations in each treatment group
- Equivalently: CRD needs about **41.9% more observations** per treatment

Practical Example:

- If RBD requires 10 observations per treatment to achieve desired precision.
- CRD would need about 14.19 observations per treatment
- In practice: Round up to 15 observations per treatment
- **Conclusion**: RBD saves about 5 observations per treatment (33% reduction)

Pairwise Comparisons of Treatments Means

The Limitations of the Overall F-test

- ANOVA F-test tells us: **Are there differences among treatments?**
- But it doesn't tell us: **Which specific treatments differ?**
- After rejecting $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$, we need more detailed analysis

Real-World Questions

- Which fertilizer gives the highest crop yield?
- Which drug is most effective?
- Which teaching method works best?
- We need **pairwise comparisons** to answer these questions

Example Scenario

Comparing 3 fertilizers (A, B, C) on crop yield:

- F-test rejects H_0: There are differences
- But is A better than B? Is B better than C? By how much?

Hypothesis Testing for Pairwise Comparisons

For a Single Pair of Treatments

For treatments i and j:

$$H_0 : \mu_i = \mu_j \quad \text{(no difference between treatments } i \text{ and } j\text{)}$$
$$H_a : \mu_i \neq \mu_j \quad \text{(treatments } i \text{ and } j \text{ differ)}$$

Equivalently: $H_0 : \tau_i = \tau_j$ vs. $H_a : \tau_i \neq \tau_j$

Multiple Comparisons Problem

With k treatments, there are $\binom{k}{2} = \frac{k(k-1)}{2}$ possible pairs:

- For $k = 3$: 3 comparisons (A vs. B, A vs. C, B vs. C)
- For $k = 4$: 6 comparisons
- Problem: Multiple tests increase Type I error rate

Family-wise Error Rate (FWER)

Probability of making **at least one** Type I error among all comparisons:

$$\text{FWER} = P(\text{Reject at least one true } H_0)$$

ANOVA Table for RBD

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{b} y_{ij}^2 - \frac{(G.T.)^2}{n}$$

$$SSTr = \sum_{i=1}^{k} \frac{T_i^2}{b} - \frac{(G.T.)^2}{n}, \text{ where } T_i = \sum_{j=1}^{b} y_{ij}$$

$$SSB = \sum_{j=1}^{b} \frac{B_j^2}{k} - \frac{(G.T.)^2}{n}, \text{ where } B_j = \sum_{i=1}^{k} y_{ij}$$

$$SSE = TSS - SSTr - SSB$$

$$\overline{T}_i = \frac{T_i}{b}, \quad \overline{B}_j = \frac{B_j}{k}, \quad n = bk; \quad d.f._{TSS} = bk - 1$$

$$d.f._{SSTr} = k - 1 \quad \Rightarrow \quad MSTr = \frac{SSTr}{k-1}, \quad d.f._{SSB} = b - 1 \quad \Rightarrow \quad MSB = \frac{SSB}{b-1}$$

$$d.f._{SSE} = (b-1)(k-1) \quad \Rightarrow \quad MSE = \frac{SSE}{(b-1)(k-1)}$$

Source	df	SS	MS	F
Treatments	$k-1$	SSTr	MSTr	$F_{\text{tr}} = \frac{MSTr}{MSE}$
Blocks	$b-1$	SSB	MSB	$F_{\text{block}} = \frac{MSB}{MSE}$
Error	$(b-1)(k-1)$	SSE	MSE	
Total	$bk-1$	TSS		

Tukey's Honestly Significant Difference (HSD)

The Problem with Individual t-tests

If we perform m independent tests each at level α:

$$\text{FWER} = 1 - (1 - \alpha)^m$$

For $\alpha = 0.05$ and $m = 6$ comparisons:

$$\text{FWER} = 1 - (1 - 0.05)^6 = 1 - 0.95^6 \approx 0.265$$

That's a 26.5% chance of at least one false positive!

Tukey's Solution

- Controls the **family-wise error rate** at α
- Simultaneous inference based on the **studentized range distribution**

Tukey's Test Statistic

$$q = \frac{|\overline{T}_i - \overline{T}_j|}{\sqrt{MSE/b}}$$

where MSE is from RBD ANOVA, b = number of blocks.

Tukey's Critical Value and Decision Rule

Critical Value

$q_\alpha(k, \nu)$ where k = # of treatments, $\nu = (b-1)(k-1)$.

Decision Rule for Pair i vs. j

Reject $H_0 : \mu_i = \mu_j$ if:

$$|\overline{T}_i - \overline{T}_j| > q_\alpha(k, \nu) \times \sqrt{\frac{MSE}{b}}$$

Honestly Significant Difference (HSD)

$$HSD = q_\alpha(k, \nu) \times \sqrt{\frac{MSE}{b}}$$

Any pair with difference $|\overline{T}_i - \overline{T}_j| > HSD$ is significantly different.

Note: The formula is the same as

$$HSD = q_\alpha(k, d.f._{SSE})\sqrt{\frac{MSE}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}.$$

Under RBD, we have $n_i = b$ for all $i = 1, \ldots, k$, so

$$\frac{MSE}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right) = \frac{MSE}{2} \cdot \frac{2}{b} = \frac{MSE}{b}.$$

Tukey's Simultaneous Confidence Intervals

For the Difference $\mu_i - \mu_j$

$$(\overline{T}_i - \overline{T}_j) \pm q_\alpha(k, \nu) \times \sqrt{\frac{MSE}{b}}$$

Properties

- **Simultaneous coverage probability**: $1 - \alpha$ for **all** intervals.
- More conservative than individual confidence intervals.

Example Interpretation

If the interval for $\mu_A - \mu_B$ is $(2.5, 5.3)$:

- Fertilizer A yields 2.5 to 5.3 kg more than Fertilizer B.
- Since 0 is not in the interval: Significant difference.

Example Fertilizer Comparison: 3 Fertilizers, 4 Farms

	Farm 1	Farm 2	Farm 3	Farm 4	T_i
Fert. A	12	14	11	13	50
Fert. B	15	16	14	17	62
Fert. C	10	12	9	11	42
B_j	37	42	34	41	$G.T. = 154$

From Previous Calculations:

$$\overline{T}_A = 12.5, \quad \overline{T}_B = 15.5, \quad \overline{T}_C = 10.5$$
$$MSE = 0.889 \quad (d.f. = 6), \quad b = 4, \quad k = 3$$

Step 1: Find Critical Value

$k = 3, \nu = 6, \alpha = 0.05 \Rightarrow q_{0.05}(3, 6) = 4.34$

Step 2: Calculate HSD

$$HSD = 4.34 \times \sqrt{\frac{0.889}{4}} = 4.34 \times 0.471 = 2.045$$

Step 3: Calculate Treatment Differences

- $|\overline{T}_A - \overline{T}_B| = |12.5 - 15.5| = 3.0$
- $|\overline{T}_A - \overline{T}_C| = |12.5 - 10.5| = 2.0$
- $|\overline{T}_B - \overline{T}_C| = |15.5 - 10.5| = 5.0$

Comparison with HSD = 2.045

- A vs. B: $3.0 > 2.045$ **Significant**
- A vs. C: $2.0 < 2.045$ **Not significant**
- B vs. C: $5.0 > 2.045$ **Significant**

Simultaneous Confidence Intervals ($\alpha = 0.05$)

$$\mu_A - \mu_B : -3.0 \pm 2.045 = (-5.045, -0.955)$$
$$\mu_A - \mu_C : 2.0 \pm 2.045 = (-0.045, 4.045)$$
$$\mu_B - \mu_C : 5.0 \pm 2.045 = (2.955, 7.045)$$

Friedman Test: A Nonparametric Alternative to RBD

Basic Idea

- Nonparametric alternative to parametric RBD ANOVA
- Tests for differences in treatment **medians** rather than means
- Based on **ranks within blocks** instead of raw data

Key Advantages

1. Does not require normality assumption
2. Robust to outliers
3. Works with ordinal data
4. Appropriate for small sample sizes

When to Use Friedman Test

- When RBD ANOVA assumptions are violated
- With ordinal or non-normal continuous data
- When data contain outliers
- With small sample sizes

Hypotheses for the Friedman Test

Null Hypothesis

H_0: All treatment medians are equal
Or equivalently:

$$H_0 : M_1 = M_2 = \cdots = M_k$$

where M_i is the population median for treatment i.

Alternative Hypothesis

H_a: At least one treatment median differs from the others
Or equivalently:

$$H_a : M_i \neq M_j \text{ for at least one pair } (i, j)$$

Testing Approach

- We test based on ranks rather than raw values
- Under H_0, each treatment should have approximately equal average rank
- Under H_a, some treatments will have systematically higher/lower ranks

Assumptions for Friedman Test

Required Assumptions:

1. **Random Sample**: Blocks are randomly selected
2. **Block Independence**: Responses within different blocks are independent
3. **Ordinal Measurement**: Data can be ranked within each block
4. **Within-Block Independence**: Within each block, observations are independent

What We DON'T Need:

- Normality of errors
- Homogeneity of variances
- Interval scale measurement (ordinal is sufficient)
- Large sample sizes

Suitable Data Types:

- Rankings within blocks (e.g., 1st, 2nd, 3rd)
- Ordinal ratings (e.g., poor, fair, good, excellent)
- Continuous data that violate normality
- Likert scale responses

Friedman Test: Ranking Procedure

Step 1: Rank Within Each Block

- For each block j, rank the k observations from 1 to k
- Assign rank 1 to the smallest value, rank 2 to the next, ..., rank k to the largest
- Handle ties by assigning average ranks

Step 2: Calculate Treatment Rank Sums

Let $T_{R_{ij}}$ be the rank of treatment i in block j, then:

$$T_{R_i} = \sum_{j=1}^{b} R_{ij} \text{ for } i = 1, 2, \ldots, k$$

where T_{R_i} represents the total rank sum for treatment i and R_{ij} denotes the blockwise rank for y_{ij} in block j. Note = $\sum_{i=1}^{k} T_{R_i} = \dfrac{bk(k+1)}{2}$.

Step 3: Calculate the Friedman Test Statistic

$$\chi_F^2 = F_R = \frac{12}{bk(k+1)} \sum_{i=1}^{k} T_{R_i}^2 - 3b(k+1)$$

Distribution Friedman Test Statistic For Large Samples (b or k large)

- Under H_0, χ^2_F approximately follows a χ^2 distribution
- Degrees of freedom: $df = k - 1$
- Critical value: $\chi^2_{\alpha,k-1}$
- Decision: Reject H_0 if $F_R = \chi^2_F > \chi^2_{\alpha,k-1}$

For Small Samples

- Use exact Friedman distribution tables
- Available for small b and k
- More accurate than χ^2 approximation

Decision Rule

Reject H_0 if $\chi^2_F \geq \chi^2_{\alpha,k-1}$

Common Critical Values ($\alpha = 0.05$)

k	$\chi^2_{0.05,k-1}$
3	5.991
4	7.815
5	9.488
6	11.070

Example: Taste Testing Experiment

Scenario:

- 4 judges (blocks) taste 3 brands of coffee (treatments)
- Rate each coffee on a scale 1-10
- Data may not be normal (subjective ratings)

Table: Coffee Taste Ratings (1-10 scale)

Judge	Brand A	Brand B	Brand C
1	7	8	6
2	6	9	5
3	8	7	4
4	7	8	5

Parametric vs. Nonparametric Approach

- Parametric: RBD ANOVA (requires normality)
- Nonparametric: Friedman test (no normality required)

Example: Ranking Within Blocks

Step 1: Rank Each Judge's Ratings

Table: Ranks Within Each Block (Judge)

Judge	T_{R_A}	T_{R_B}	T_{R_C}	Note
1	2	3	1	7(2), 8(3), 6(1)
2	2	3	1	6(2), 9(3), 5(1)
3	3	2	1	8(3), 7(2), 4(1)
4	2	3	1	7(2), 8(3), 5(1)
T_{R_i}	9	11	4	Rank sums

Step 2: Calculate Rank Sums

$$T_{R_A} = 2 + 2 + 3 + 2 = 9$$
$$T_{R_B} = 3 + 3 + 2 + 3 = 11$$
$$T_{R_C} = 1 + 1 + 1 + 1 = 4$$

Step 3: Check Mean Rank Under H_0, expected rank sum for each treatment:

$$E[T_{R_i}] = \frac{b(k+1)}{2} = \frac{4 \times (3+1)}{2} = \frac{16}{2} = 8$$

Example: Calculating Friedman Test Statistic (Cont.)

Given:

- $b = 4$ blocks (judges)
- $k = 3$ treatments (coffee brands)
- Rank sums: $T_{R_A} = 9, T_{R_B} = 11, T_{R_C} = 4$

Calculate χ^2_F:

$$\chi^2_F = \frac{12}{bk(k+1)} \sum_{i=1}^{k} T^2_{R_i} - 3b(k+1)$$

$$\sum_{i=1}^{k} T^2_{R_i} = 9^2 + 11^2 + 4^2 = 81 + 121 + 16 = 218$$

$$\chi^2_F = \frac{12}{4 \times 3 \times 4} \times 218 - 3 \times 4 \times 4$$

$$= \frac{12}{48} \times 218 - 48 = 0.25 \times 218 - 48 = 54.5 - 48 = 6.5$$

Example: Hypothesis Testing Decision (Cont.)

Test Statistic:

$\chi^2_F = 6.5$

Critical Value ($\alpha = 0.05$, $df = k - 1 = 2$):

$\chi^2_{0.05,2} = 5.991$

Decision:

Since $\chi^2_F = 6.5 > 5.991 = \chi^2_{0.05,2}$, we reject H_0.

Conclusion:

There is sufficient evidence to conclude that at least one coffee brand has a different median rating from the others at the 0.05 significance level.

Note:

The Friedman test doesn't tell us **which** brands differ - for that we need multiple comparisons (Nemenyi's procedure).

Nemenyi's Procedure: Nonparametric Multiple Comparisons

Purpose: After rejecting H_0 in Friedman test, identify which specific treatments differ.

Procedure

1. Calculate mean ranks: $\overline{R_i} = \frac{T_{R_i}}{b}$
2. Determine critical difference (Ne) using studentized range distribution
3. Compare absolute differences in mean ranks to Ne
4. Construct simultaneous confidence intervals for differences in mean ranks

Critical Difference Formula

$$Ne = q_\alpha(k, \infty) \times \sqrt{\frac{k(k+1)}{12b}}$$

where:

- $q_\alpha(k, \infty)$ = critical value from studentized range distribution
- k = number of treatments, and b = number of blocks

Decision Rule

Treatments i and j differ significantly if:

$$|\overline{R_i} - \overline{R_j}| > Ne$$

Nemenyi's Simultaneous Confidence Intervals

For the Difference in Mean Ranks $\bar{T}_{R_i} - \bar{T}_{R_j}$

$$(\overline{R_i} - \overline{R_j}) \pm Ne$$

where $Ne = q_\alpha(k, \infty) \times \sqrt{\frac{k(k+1)}{12b}}$

Properties of Simultaneous Confidence Intervals

- **Simultaneous coverage probability**: $1 - \alpha$
- Probability that **all** intervals contain their true differences is $1 - \alpha$
- Accounts for multiple comparisons
- If interval does not contain 0, treatments differ significantly

Interpretation

With $100(1 - \alpha)\%$ confidence, **all** true differences in mean ranks lie within their respective intervals.

Relationship to Hypothesis Testing

- Reject $H_0 : \overline{R_i} = \overline{R_j}$ if 0 is not in the interval
- Fail to reject H_0 if 0 is in the interval

Nemenyi's Procedure for Coffee Example

From Previous Calculations:

- $b = 4, k = 3$
- Rank sums: $T_{R_A} = 9, T_{R_B} = 11, T_{R_C} = 4$
- Mean ranks: $\overline{R_A} = 2.25, \overline{R_B} = 2.75, \overline{R_C} = 1.00$

Step 1: Find Critical Value

For $\alpha = 0.05, k = 3, \infty$ df: $q_{0.05}(3, \infty) = 3.314$

Step 2: Calculate Critical Difference

$$Ne = 3.314 \times \sqrt{\frac{3(3+1)}{12 \times 4}} = 3.314 \times \sqrt{\frac{12}{48}}$$

$$Ne = 3.314 \times \sqrt{0.25} = 3.314 \times 0.5 = 1.657$$

Nemenyi's Procedure: Pairwise Comparisons

Pairwise Differences in Mean Ranks:

- $|\overline{R_A} - \overline{R_B}| = |2.25 - 2.75| = 0.50$
- $|\overline{R_A} - \overline{R_C}| = |2.25 - 1.00| = 1.25$
- $|\overline{R_B} - \overline{R_C}| = |2.75 - 1.00| = 1.75$

Compare with Critical Difference ($Ne = 1.657$):

- A vs. B: $0.50 < 1.657$ **Not significant**
- A vs. C: $1.25 < 1.657$ **Not significant**
- B vs. C: $1.75 > 1.657$ **Significant**

Simultaneous Confidence Intervals ($\alpha = 0.05$):

- A - B: $-0.50 \pm 1.657 = (-2.157, 1.157)$
- A - C: $1.25 \pm 1.657 = (-0.407, 2.907)$
- B - C: $1.75 \pm 1.657 = (0.093, 3.407)$

Interpretation of Results

Hypothesis Testing Conclusion:

- Only Brand B and Brand C differ significantly
- Brand A is not significantly different from either B or C
- Brand B has the highest mean rank (2.75), Brand C the lowest (1.00)

Confidence Interval Interpretation:

- A - B: CI contains 0 $\Rightarrow$ Not significant
- A - C: CI contains 0 $\Rightarrow$ Not significant
- B - C: CI does not contain 0 $\Rightarrow$ Significant

Practical Interpretation:

- With 95% confidence (family-wise), Brand B has a higher mean rank than Brand C
- The true difference in mean ranks between B and C is between 0.093 and 3.407
- Brands A and B, and A and C cannot be distinguished based on this data

Parametric vs. Nonparametric RBD Analysis

Table: Comparison of Parametric RBD ANOVA and Friedman Test

Parametric RBD ANOVA	**Friedman Test**
Tests differences in **means**	Tests differences in **medians**
Requires **normality**	No normality required
Requires **homogeneous variances**	No homogeneity required
Uses **raw data**	Uses **ranks within blocks**
Test statistic: F	Test statistic: $F_R = \chi^2_F$
Multiple comparisons: **Tukey's HSD**	Multiple comparisons: **Nemenyi's**
More **powerful** if assumptions met	More **robust** if assumptions violated

Choose Parametric RBD ANOVA when:

- Data are approximately normal
- Sample sizes are large (CLT applies)
- Variances are homogeneous

Choose Friedman Test when:

- Data are not normally distributed
- Data are ordinal (rankings, Likert scales)
- There are outliers present